

CIA-1-QB

Set A

Part-A

1. Draw the block diagram of communication systems.
2. Compare DSB-SC and SSB.
3. For an FM system, the maximum deviation is 75 kHz and the highest modulating frequency is 15 kHz. Determine the required transmission bandwidth using Carson's rule.
4. Identify suitable modulation technique for TV transmission and justify.
5. Classify the FM signal as NBFM or WBFM if $\beta = 0.5$.
6. Summarize the key differences between a random variable and a random process.
7. Discuss the mathematical conditions for Wide Sense Stationary.
8. Show how narrowband noise behaves in a modulated system and state its characteristics.
9. Apply the noise equivalent bandwidth formula to analyze receiver performance.

Part-B

10. Q10

1. Demonstrate how an AM signal is generated using a square law modulator and derive its mathematical equation
2. Construct the block diagram based on Weaver's method and describe how it generates an SSB signal.

11. Q11

1. Apply angle modulation principles to formulate the mathematical expressions of FM and PM waves.
2. Illustrate the process of FM demodulation with balanced slope detector with the circuit diagram and characteristic curve

12. Q12

1. Illustrate the terms mean, correlation, covariance and ergodicity.
2. Interpret the process of autocorrelation and explain the properties of autocorrelation function.

13. Q13

1. Apply the theory of internal and external noise sources to analyze their impact on communication systems.
2. Demonstrate how noise temperature is related to noise figure by applying SNR concepts.

Set B

Part-A

1. Infer the need for modulation in communication systems.
2. Write and explain the general expression for an AM signal.
3. Identify whether distortion occurs if $\mu > 1$.

4. Using Carson's rule, calculate bandwidth if $\Delta f = 75$ kHz and $f_m = 15$ kHz.
5. Classify the FM signal as NBFM or WBFM if $\beta = 8$.
6. Summarize the important properties of ergodic and Gaussian random processes.
7. Outline the essential properties that an autocorrelation function must satisfy.
8. Apply the relationship between noise figure and noise temperature to analyze receiver characteristics.
9. Demonstrate how thermal agitation in a resistor produces noise voltage and derive its mathematical expression.

Part-B

10. Q10

1. Construct the balanced modulator configuration for DSB-SC generation and describe its operation.
2. Apply the concept of vestigial sideband modulation to design a transmitter and receiver block diagram and explain their operation.

11. Q11

1. Show how Wideband FM is expressed mathematically and evaluate its performance characteristics relative to Narrowband FM.
2. Construct the phasor representation of voltages in a Foster-Seeley discriminator and explain how it recovers the modulating signal.

12. Q12

1. Formulate the mathematical representation of a random process using statistical parameters.
2. Show how the statistical properties determine whether a process is WSS or SSS and compare them.

13. Q13

1. Show how noise figure influences overall receiver sensitivity and performance.
2. Demonstrate how narrowband noise is represented and analyze the behavior of its in-phase and quadrature components.